Microsoft Azure

Complete Textbook

Cloud Computing Fundamentals to Advanced Architecture

Azure Cloud Platform

```
┌──────────────────────────────────────────────────┐
│          MICROSOFT AZURE GLOBAL CLOUD            │
│                                                  │
│   Compute   Storage   Network   AI/ML   Security │
│                                                  │
│      60+ Regions • 200+ Services • 99.99% SLA    │
└──────────────────────────────────────────────────┘
```

Covers AZ-900 • AZ-104 • AZ-204 • AZ-305 Exam Objectives

Version 2025 Edition

Chapter 1: Introduction to Cloud Computing & Microsoft Azure

1.1 What is Cloud Computing?

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ('the cloud') to offer faster innovation, flexible resources, and economies of scale. Organizations pay only for the cloud services they use, which helps lower operating costs, run infrastructure more efficiently, and scale as business needs change.

> **Key Definition**
>
> Cloud Computing: On-demand availability of computer system resources—especially data storage and computing power—without direct active management by the user.
>
> Pay-as-you-go: You only pay for the services you consume, eliminating the need for large upfront capital expenditure.

1.2 Cloud Service Models

Cloud computing is offered in three primary service models, each providing different levels of abstraction and management responsibility.

```
            Cloud Service Models – Shared Responsibility

  On-Premises      IaaS           PaaS           SaaS
  ┌──────────┐   ┌──────────┐   ┌──────────┐   ┌──────────┐
  │ Apps     │   │ Apps     │   │ Apps     │   │▓▓▓▓▓▓▓▓▓▓│
  │ Runtime  │   │ Runtime  │   │▓▓▓▓▓▓▓▓▓▓│   │▓▓▓▓▓▓▓▓▓▓│
  │ OS       │   │ OS       │   │▓▓▓▓▓▓▓▓▓▓│   │▓▓▓▓▓▓▓▓▓▓│
  │ Hardware │   │▓▓▓▓▓▓▓▓▓▓│   │▓▓▓▓▓▓▓▓▓▓│   │▓▓▓▓▓▓▓▓▓▓│
  └──────────┘   └──────────┘   └──────────┘   └──────────┘
  You manage       You  ▓ = Provider manages
```

Model	Description
IaaS	Infrastructure as a Service – You manage OS, middleware, runtime, apps, data. Provider manages virtualization, servers, storage, networking.
PaaS	Platform as a Service – Provider manages infrastructure + OS + runtime. You manage only your applications and data.
SaaS	Software as a Service – Fully managed by provider. Users access software via browser (e.g., Microsoft 365, Dynamics 365).

FaaS	Function as a Service / Serverless – Execute code without managing servers. Azure Functions is Microsoft's offering.

1.3 Cloud Deployment Models

Model	Infrastructure	Control	Use Case
Public Cloud	Shared, multi-tenant	Low	Startups, web apps
Private Cloud	Dedicated, single-tenant	High	Regulated industries
Hybrid Cloud	Mix of both	Medium	Gradual migration
Multi-Cloud	Multiple providers	Variable	Vendor independence

1.4 Benefits of Cloud Computing

- High Availability & Reliability: SLAs guarantee uptime (Azure offers 99.99% for many services)
- Scalability: Instantly scale out (add instances) or scale up (increase instance size)
- Elasticity: Automatically add/remove resources based on demand
- Agility: Deploy resources in minutes rather than weeks
- Geo-distribution: Deploy applications to global datacenters for low latency
- Disaster Recovery: Built-in redundancy and backup capabilities
- Cost Savings: Convert CapEx to OpEx; pay only for what you use
- Security: Microsoft invests $1 billion+ annually in cybersecurity

1.5 Microsoft Azure Overview

Microsoft Azure, commonly referred to as Azure, is a cloud computing service created by Microsoft for building, testing, deploying, and managing applications and services through Microsoft-managed data centers. Azure provides software as a service (SaaS), platform as a service (PaaS) and infrastructure as a service (IaaS) and supports many different programming languages, tools, and frameworks.

Azure Global Infrastructure

```
┌─────────────────────────────────────────────────────────┐
│                    Azure Geography                      │
│  ┌──────────────┐     ┌──────────────┐                 │
│  │   Region     │     │   Region     │  (60+ worldwide) │
│  │  ┌────────┐  │     │  ┌────────┐  │                 │
│  │  │ Zone1  │  │     │  │ Zone1  │  │                 │
│  │  │ Zone2  │  │     │  │ Zone2  │  │                 │
│  │  │ Zone3  │  │     │  │ Zone3  │  │                 │
│  │  └────────┘  │     │  └────────┘  │                 │
│  └──────────────┘     └──────────────┘                 │
```

```
│            Connected via Azure Backbone Network            │
└────────────────────────────────────────────────────────────┘
```

Term	Definition
Geography	A market (e.g., United States, Europe) with 2+ regions for data residency compliance
Region	A set of datacenters deployed within a latency-defined perimeter
Region Pair	Each region is paired with another region in the same geography for disaster recovery
Availability Zone	Physically separate datacenters within a region with independent power/cooling/networking
Edge Zone	Extended Azure compute at carrier locations for ultra-low latency

Chapter 2: Azure Core Services – Compute

Azure Compute is the foundation of cloud workloads. It encompasses virtual machines, containers, serverless functions, and managed application hosting. Understanding compute options allows architects to choose the right service for each workload type.

2.1 Azure Virtual Machines (VMs)

Azure Virtual Machines provide on-demand, scalable computing resources that give you the flexibility of virtualization without having to buy and maintain the physical hardware. VMs are ideal for lift-and-shift migrations and workloads requiring full OS control.

Azure VM Architecture

```
┌─────────────────────────────────────────────────────┐
│               Azure Subscription                    │
│  ┌──────────────────────────────────────────────┐   │
│  │         Resource Group                        │  │
│  │  ┌───────────────┐  ┌───────────────────────┐ │  │
│  │  │  Virtual      │  │  Network Interface    │ │  │
│  │  │  Machine      │  │  Card (NIC)           │ │  │
│  │  │  ┌─────────┐  │  │  ┌──────────────┐    │ │  │
│  │  │  │OS Disk  │  │  │  │Private IP    │    │ │  │
│  │  │  │Data Disk│  │  │  │Public IP     │    │ │  │
│  │  │  └─────────┘  │  │  └──────────────┘    │ │  │
│  │  └───────────────┘  └───────────────────────┘ │  │
│  └──────────────────────────────────────────────┘   │
└─────────────────────────────────────────────────────┘
```

VM Series & Use Cases

Series	Name	vCPUs	RAM	Best For
B	Burstable	1-20	0.5-80 GB	Dev/Test, low-traffic
D	General Purpose	2-96	8-384 GB	Enterprise apps, DBs
E	Memory Optimized	2-96	16-672 GB	In-memory databases
F	Compute Optimized	2-72	4-144 GB	Batch, gaming, web
H	High Performance	8-80	32-2TB	HPC simulations
N	GPU Optimized	6-64	56-440 GB	AI/ML, rendering

L	Storage Optimized	8-80	64-640 GB	NoSQL, big data

2.2 VM Scale Sets & Availability

Azure Virtual Machine Scale Sets let you create and manage a group of load-balanced VMs. The number of VM instances can automatically increase or decrease in response to demand or a defined schedule.

VM Scale Set with Load Balancer

```
Internet
   |
┌──▼──────────────────┐
│   Azure Load        │
│   Balancer          │
└──┬─────────┬────────┘
   |         |         |
┌──▼──┐  ┌───▼──┐  ┌───▼──┐
│ VM1 │  │ VM2  │  │ VM3  │  ◄── Auto-scaled
└─────┘  └──────┘  └──────┘
   |         |         |
┌──▼─────────▼─────────▼──┐
│     Shared Data Store    │ |
└─────────────────────────┘
```

Concept	Description
Availability Set	Protects VMs from hardware failures within a datacenter using fault domains & update domains
Availability Zone	Deploys VMs across physically separate zones within a region for zone-level protection
Scale Set	Automatically manages and scales identical VM instances based on rules or schedule
Spot VMs	Up to 90% discounted VMs using Azure's spare capacity (can be evicted with 30s notice)

2.3 Azure App Service

Azure App Service is a fully managed platform for building, deploying, and scaling web apps, REST APIs, and mobile backends. It supports .NET, Java, Ruby, Node.js, PHP, and Python. You focus on code; Azure handles infrastructure.

App Service Plan Tiers

Free/Shared: Dev/test only – shared infrastructure, no SLA

> Basic: Dedicated VMs, manual scale, dev/test
> Standard: Auto-scale, custom domains, SSL certificates, SLA 99.95%
> Premium: Enhanced performance, VNet integration, more instances
> Isolated: Fully isolated, private environment in Azure VNet (App Service Environment)

2.4 Azure Kubernetes Service (AKS)

Azure Kubernetes Service simplifies deploying, managing, and scaling containerized applications using Kubernetes. Azure handles critical tasks like health monitoring and maintenance. AKS is ideal for microservices architectures and cloud-native applications.

AKS Cluster Architecture

```
┌───────────────────────────────────────────────────────┐
│                   AKS Cluster                        │
│  ┌───────────────────────────────────────────────┐   │
│  │                Control Plane (managed by Azure) │ │
│  │  kube-apiserver │ etcd │ kube-scheduler       │ │
│  └───────────────────────────────────────────────┘ │
│                         │                           │
│  ┌─────────┐  ┌─────────┐  ┌─────────┐             │
│  │ Node 1  │  │ Node 2  │  │ Node 3  │  Node Pool  │
│  │ ┌─────┐ │  │ ┌─────┐ │  │ ┌─────┐ │             │
│  │ │Pod A│ │  │ │Pod C│ │  │ │Pod E│ │             │
│  │ │Pod B│ │  │ │Pod D│ │  │ │Pod F│ │             │
│  │ └─────┘ │  │ └─────┘ │  │ └─────┘ │             │
│  └─────────┘  └─────────┘  └─────────┘             │
└───────────────────────────────────────────────────────┘
```

2.5 Azure Functions (Serverless)

Azure Functions is a serverless compute service that lets you run event-triggered code without provisioning or managing servers. It supports triggers from HTTP, timers, Azure Storage, Event Hubs, Service Bus, Cosmos DB, and more.

Azure Functions – Event-Driven Flow

```
Trigger            Function          Output Binding
┌───────────┐      ┌───────────┐      ┌──────────────┐
│HTTP Req   │────▶│           │────▶│ Cosmos DB    │
│Timer      │────▶│  Your     │────▶│ Service Bus  │
│Blob Event │────▶│  Code     │────▶│ SendGrid     │
│Event Hub  │────▶│  (auto-   │────▶│ Blob Storage │
│Queue Msg  │────▶│  scaled)  │────▶│ SignalR      │
└───────────┘      └───────────┘      └──────────────┘
```

Chapter 3: Azure Storage Services

Azure Storage is Microsoft's cloud storage solution providing highly available, massively scalable, durable, and secure storage for a variety of data objects. Azure Storage offers five core services: Blob Storage, Azure Files, Queue Storage, Table Storage, and Azure Disk Storage.

3.1 Azure Storage Account

An Azure Storage Account is the top-level resource for Azure Storage. It provides a unique namespace in Azure for your data, and every object you store in Azure Storage has an address that includes your unique account name.

Storage Account Hierarchy

```
┌─────────────────────────────────────────────────────┐
│             Storage Account                         │
│  myaccount.blob.core.windows.net                    │
│  myaccount.file.core.windows.net                    │
│  myaccount.queue.core.windows.net                   │
│  myaccount.table.core.windows.net                   │
│                                                     │
│  ┌──────────┐  ┌─────────┐  ┌───────┐  ┌───────┐  │
│  │  Blob    │  │  Files  │  │ Queue │  │ Table │  │
│  │ Storage  │  │  Share  │  │       │  │       │  │
│  │Container │  │         │  │       │  │       │  │
│  └──────────┘  └─────────┘  └───────┘  └───────┘  │
└─────────────────────────────────────────────────────┘
```

Storage Account Types

Type	Supported Services	Performance	Redundancy
Standard GPv2	Blob, Files, Queue, Table	Standard (HDD)	LRS/GRS/ZRS/GZRS
Premium Block Blobs	Block & Append Blobs	Premium (SSD)	LRS/ZRS
Premium File Shares	Azure Files only	Premium (SSD)	LRS/ZRS
Premium Page Blobs	Page Blobs only	Premium (SSD)	LRS only

3.2 Azure Blob Storage

Azure Blob Storage is optimized for storing massive amounts of unstructured data—text or binary data—such as images, videos, documents, backups, and logs. Blob stands for Binary Large Object.

Blob Type	Use Case
Block Blob	Store text and binary data up to 4.75 TB. Ideal for images, documents, and media files
Append Blob	Optimized for append operations. Best for logging and auditing scenarios
Page Blob	Store random-access files up to 8 TB. Foundation for Azure VM disks (VHDs)

Blob Access Tiers

Tier	Access Frequency	Storage Cost	Access Cost	Minimum Duration
Hot	Frequent	Highest	Lowest	None
Cool	Infrequent	Lower	Higher	30 days
Cold	Rare	Even Lower	Even Higher	90 days
Archive	Very rare	Lowest	Highest	180 days

> **Lifecycle Management**
> Azure Blob Storage lifecycle management lets you define rules to automatically transition blobs between access tiers or delete them based on age or last modification time.
> Example: Move to Cool after 30 days → Archive after 90 days → Delete after 365 days.

3.3 Azure Files

Azure Files offers fully managed file shares in the cloud that are accessible via the industry-standard Server Message Block (SMB) protocol and Network File System (NFS) protocol. Azure file shares can be mounted concurrently by cloud or on-premises deployments of Windows, Linux, and macOS.

3.4 Storage Redundancy Options

```
           Storage Redundancy — LRS vs ZRS vs GRS vs GZRS

   LRS (Local)        ZRS (Zone)          GRS (Geo)
   ┌───────────┐     ┌──────────────┐     ┌──────┐  ┌──────┐
   │ Region    │     │ Region       │     │Region│  │Region│
   │ ┌─┐  ┌─┐  │     │ ┌──┐ ┌──┐ ┌──┐│    │ ┌──┐ │  │ ┌──┐ │
   │ │C│  │C│  │     │ │Z1│ │Z2│ │Z3││    │ │3x│ │  │ │3x│ │
   │ │o│  │o│  │     │ └──┘ └──┘ └──┘│    │ └──┘ │  │ └──┘ │
   │ │p│  │p│  │     │ 3 copies      │    │Primary│ │Second│
   │ └─┘  └─┘  │     │ in 3 zones    │    └──────┘  └──────┘
```

```
   │3 copies    │     └───────────────┘    Async replication
   └────────────┘
```

Chapter 4: Azure Networking

Azure Networking forms the backbone connecting all Azure services and linking Azure environments to on-premises datacenters and the internet. Understanding networking is critical for building secure, scalable, and highly available architectures.

4.1 Azure Virtual Network (VNet)

Azure Virtual Network is the fundamental building block for private networking in Azure. A VNet enables many Azure resources, such as Azure Virtual Machines, to securely communicate with each other, the internet, and on-premises networks.

VNet Architecture with Subnets

```
┌─────────────────────────────────────────────────────────┐
│              Virtual Network: 10.0.0.0/16               │
│                                                         │
│  ┌──────────────────────┐  ┌──────────────────────┐     │
│  │ Web Subnet           │  │ App Subnet           │     │
│  │ 10.0.1.0/24          │  │ 10.0.2.0/24          │     │
│  │  ┌──────┐ ┌──────┐   │  │  ┌──────┐ ┌──────┐   │     │
│  │  │ VM1  │ │ VM2  │   │  │  │ VM3  │ │ VM4  │   │     │
│  │  └──────┘ └──────┘   │  │  └──────┘ └──────┘   │     │
│  │ NSG: Allow 80/443    │  │ NSG: Allow 8080      │     │
│  └──────────────────────┘  └──────────────────────┘     │
│                                                         │
│  ┌──────────────────────┐  ┌──────────────────────┐     │
│  │ DB Subnet            │  │ Gateway Subnet       │     │
│  │ 10.0.3.0/24          │  │ 10.0.255.0/27        │     │
│  │ NSG: Allow 1433      │  │ VPN/ExpressRoute GW  │     │
│  └──────────────────────┘  └──────────────────────┘     │
└─────────────────────────────────────────────────────────┘
```

4.2 Network Security Groups (NSG)

A Network Security Group contains security rules that allow or deny inbound or outbound network traffic to Azure resources. Each rule specifies source, destination, port, and protocol.

NSG Property	Details
Priority	Rules are processed in priority order (100-4096); lower number = higher priority
Source/Dest	IP address, CIDR range, service tag, or application security group
Port	Specific port number or range (e.g., 80, 443, 3389, 1024-65535)

Protocol	TCP, UDP, ICMP, or Any
Action	Allow or Deny
Service Tags	Simplified IP prefixes for Azure services (e.g., AzureLoadBalancer, Internet, VirtualNetwork)

4.3 Azure Load Balancer

Load Balancer Types

```
Public Load Balancer           Internal Load Balancer
┌────────────────────┐         ┌─────────────────────┐
│   Internet         │         │    VNet Internal     │
│       │            │         │         │            │
│   ┌───▼───┐        │         │    ┌────▼────┐       │
│   │  LB   │  Pub   │         │    │   ILB   │  Priv │
│   │Frontend│ IP    │         │    │ Frontend│  IP   │
│   └───┬───┘        │         │    └────┬────┘       │
│   ┌───┴────────┐   │         │    ┌────┴────────┐    │
│   │Backend Pool│   │         │    │ Backend Pool│    │
│   │ VM1 VM2 VM3│   │         │    │ VM1  VM2  VM3│   │
│   └────────────┘   │         │    └─────────────┘    │
└────────────────────┘         └─────────────────────┘
```

4.4 Azure Application Gateway & WAF

Azure Application Gateway is a web traffic load balancer that enables you to manage traffic to your web applications at the application layer (Layer 7). It provides SSL/TLS termination, cookie-based session affinity, URL-based routing, and Web Application Firewall (WAF) capabilities.

4.5 Azure DNS

Azure DNS is a hosting service for DNS domains that provides name resolution using Microsoft Azure infrastructure. By hosting your domains in Azure, you can manage your DNS records using the same credentials, APIs, tools, and billing as your other Azure services.

4.6 Hybrid Connectivity

Hybrid Network Connectivity Options

```
On-Premises                        Azure
┌───────────────┐                  ┌──────────────────────┐
│  Corporate    │──VPN Gateway──▶──│  Azure VNet          │
│  Network      │                  │                      │
```

```
│              │──ExpressRoute──►─│  Private connection │
└──────────────┘                  └─────────────────────┘

VPN Gateway: Encrypted tunnel over internet (IPSec/IKE)
ExpressRoute: Private dedicated circuit (up to 100 Gbps)
VPN + ExpressRoute: ExpressRoute with VPN failover
```

Chapter 5: Azure Identity & Security

Security is a top priority in Azure. Microsoft employs a shared responsibility model where both Microsoft and the customer are responsible for different layers of security. Azure provides a comprehensive set of identity, access management, and security services.

5.1 Azure Active Directory (Azure AD / Entra ID)

Microsoft Entra ID (formerly Azure Active Directory) is Microsoft's cloud-based identity and access management service. It helps employees sign in and access resources in external resources (Microsoft 365, Azure portal) and internal resources (apps on corporate network).

Entra ID – Identity Flow

```
┌────────────┐   Authentication   ┌───────────────────────┐
│   User     │───────────────────▶│  Microsoft Entra ID   │
│ (Browser/  │                    │  ┌─────────────────┐  │
│  App)      │◀─ Token ───────────│  │ Identity Store  │  │
└────────────┘                    │  │ Policies        │  │
      │                           │  │ MFA Settings    │  │
      │ Access Token              │  └─────────────────┘  │
      │                           └───────────────────────┘
      ▼
┌─────────────────────────────────────────────────────────┐
│  Protected Resources (Azure, M365, SaaS Apps, APIs)     │
└─────────────────────────────────────────────────────────┘
```

Concept	Definition
Tenant	A dedicated instance of Entra ID representing an organization
User	An identity (person or service) managed in Entra ID
Group	Collection of users for simplified access management
Application	A registered app that can use Entra ID for authentication
Service Principal	Security identity used by applications and services (non-human)
Managed Identity	Auto-managed service principal for Azure resources (no credentials to manage)

5.2 Role-Based Access Control (RBAC)

Azure RBAC is an authorization system that provides fine-grained access management to Azure resources. It allows you to grant only the amount of access that users need to perform their jobs (principle of least privilege).

```
                         RBAC Model

  Security Principal    Role Definition     Scope
 ┌────────────────┐    ┌───────────────┐   ┌───────────────┐
 │ User           │    │ Owner         │   │ Management    │
 │ Group          │  + │ Contributor   │ + │ Group         │
 │ Service Princ  │    │ Reader        │   │ Subscription  │
 │ Managed Ident  │    │ Custom Role   │   │ Res. Group    │
 └────────────────┘    └───────────────┘   │ Resource      │
                                            └───────────────┘
               = Role Assignment
```

5.3 Azure Key Vault

Azure Key Vault is a cloud service for securely storing and accessing secrets, encryption keys, and certificates. It helps eliminate the need to store sensitive information in code and centralizes secret management.

Key Vault – What it Stores
Secrets: Passwords, database connection strings, API keys, tokens
Keys: Cryptographic keys for data encryption (RSA, EC keys; HSM-backed available)
Certificates: SSL/TLS certificates with automatic renewal support
Access Control: Integrated with Entra ID and RBAC for fine-grained access

5.4 Microsoft Defender for Cloud

Microsoft Defender for Cloud is a cloud security posture management (CSPM) and cloud workload protection platform (CWPP). It finds weaknesses in your configurations, helps strengthen your security posture, and protects workloads from evolving threats.

5.5 Zero Trust Security Model

The Zero Trust model assumes breach and verifies each request as though it originates from an open network. Rather than trust everything inside a corporate network, Zero Trust requires verification of every user, device, and connection.

```
                    Zero Trust Principles

 ┌─────────────────────────────────────────────────────┐
 │             Zero Trust Security Model               │
 │                                                     │
```

```
│  1. VERIFY EXPLICITLY                               │
│     Always authenticate and authorize based on      │
│     all available data points                       │
│                                                     │
│  2. USE LEAST PRIVILEGE ACCESS                      │
│     Limit user access with JIT/JEA policies         │
│                                                     │
│  3. ASSUME BREACH                                   │
│     Minimize blast radius, segment access,          │
│     verify end-to-end encryption                    │
└─────────────────────────────────────────────────────┘
```

Chapter 6: Azure Databases & Data Services

Azure provides a comprehensive portfolio of fully managed database services covering relational, NoSQL, in-memory, and analytical data stores. These services handle patching, backups, high availability, and scaling automatically.

6.1 Azure SQL Database

Azure SQL Database is a fully managed relational database service based on the latest stable version of Microsoft SQL Server Database Engine. It provides built-in high availability, automated backups, intelligent performance optimization, and scalability.

Feature	SQL Database	SQL Managed Instance	SQL on VM
Deployment	PaaS	PaaS	IaaS
SQL Server Compat	~99%	~100%	100%
HA Built-in	Yes	Yes	Manual
Auto Backups	Yes	Yes	Manual
Cross-DB Queries	No	Yes	Yes
Linked Servers	No	Yes	Yes
Best For	New cloud apps	Lift-and-shift	Full control

6.2 Azure Cosmos DB

Azure Cosmos DB is a fully managed, globally distributed, multi-model database service. It guarantees single-digit millisecond response times at any scale and offers multiple consistency levels, making it ideal for globally distributed applications.

Cosmos DB – Global Distribution

```
┌──────────────────────────────────────────────────────┐
│           Cosmos DB Account (Global)                 │
│                                                      │
│  ┌─────────┐   ┌─────────┐   ┌─────────┐            │
│  │  East   │   │  West   │   │  North  │            │
│  │  US     │◄─►│  Europe │◄─►│ Europe  │            │
│  │ Write   │   │ Read    │   │ Read    │            │
│  │ Region  │   │ Replica │   │ Replica │            │
│  └─────────┘   └─────────┘   └─────────┘            │
│       │                                              │
│  Multi-master: all regions can accept writes         │
```

```
│  <10ms reads and writes at 99th percentile          │
└─────────────────────────────────────────────────────┘
```

Cosmos DB APIs

API	Description
NoSQL (Core SQL)	Document model; SQL-like query syntax; JSON documents
MongoDB	Wire-compatible with MongoDB; migrate existing MongoDB apps
Cassandra	Column-family store; CQL query language; migrate from Cassandra
Gremlin	Graph database; store entities as vertices and relationships as edges
Table	Key-value store; drop-in replacement for Azure Table Storage
PostgreSQL	Distributed PostgreSQL powered by Citus (horizontal sharding)

Cosmos DB Consistency Levels

Consistency	Availability	Latency	Ordering	Reads
Strong	Lower	Higher	Guaranteed	Latest
Bounded Staleness	High	Medium	Guaranteed	Lagged
Session	High	Low	Prefix guaranteed	Your writes
Consistent Prefix	Highest	Low	Partial	Ordered
Eventual	Highest	Lowest	None	Stale possible

6.3 Azure Database for PostgreSQL & MySQL

Azure provides fully managed services for both PostgreSQL and MySQL, offering built-in high availability, automated backups with up to 35 days point-in-time restore, and intelligent performance recommendations.

6.4 Azure Synapse Analytics

Azure Synapse Analytics is an unlimited analytics service that brings together enterprise data warehousing and Big Data analytics. It gives you the freedom to query data on your terms, at scale, using either serverless or dedicated resources.

```
Azure Synapse Architecture
```

```
┌──────────────────────────────────────────────────────────┐
│                 Azure Synapse Analytics                  │
│                                                         │
│  ┌────────────┐  ┌────────────┐  ┌───────────────────┐  │
│  │ Synapse    │  │ Apache     │  │ Synapse Link      │  │
│  │ SQL Pool   │  │ Spark Pool │  │ (Cosmos DB,       │  │
│  │(dedicated/ │  │  (Big Data │  │  Dataverse)       │  │
│  │ serverless)│  │  & ML)     │  └───────────────────┘  │
│  └──────┬─────┘  └──────┬─────┘                         │
│         └───────┬───────┘                               │
│         ┌───────▼─────────┐                             │
│         │  Azure Data     │                             │
│         │  Lake Gen2      │                             │
│         └─────────────────┘                             │
│                                                         │
└──────────────────────────────────────────────────────────┘
```

Chapter 7: Azure AI & Machine Learning

Azure provides a comprehensive AI platform that enables developers and data scientists to build, deploy, and scale AI models and applications. From pre-built cognitive APIs to custom ML model training, Azure covers the full AI lifecycle.

7.1 Azure AI Services (Cognitive Services)

Azure AI Services (formerly Cognitive Services) provides pre-built AI capabilities through simple REST APIs, enabling developers to add intelligent features without AI expertise.

Azure AI Services Categories

```
┌──────────────────────────────────────────────────────────┐
│                  Azure AI Services                        │
│                                                          │
│  ┌───────────┐  ┌───────────┐  ┌─────────┐  ┌─────────┐  │
│  │  Vision   │  │ Language  │  │ Speech  │  │Decision │  │
│  │           │  │           │  │         │  │         │  │
│  │ Computer  │  │ Text      │  │Speech-to│  │Anomaly  │  │
│  │ Vision    │  │ Analytics │  │  -Text  │  │Detector │  │
│  │ Face API  │  │ Translator│  │Text-to- │  │Content  │  │
│  │ Custom    │  │ LUIS      │  │  Speech │  │Moderator│  │
│  │ Vision    │  │ QnA Maker │  │Speaker  │  │Personaliz│ │
│  └───────────┘  └───────────┘  └─────────┘  └─────────┘  │
└──────────────────────────────────────────────────────────┘
```

7.2 Azure Machine Learning

Azure Machine Learning is an enterprise-grade machine learning service for building and deploying ML models faster. It provides AutoML, MLOps pipelines, model registry, and deployment to multiple targets.

Azure ML Lifecycle

```
┌──────────┐    ┌──────────┐    ┌──────────┐    ┌──────────┐
│   Data   │───▶│  Train   │───▶│  Deploy  │───▶│ Monitor  │
│          │    │          │    │          │    │          │
│ Datasets │    │ AutoML   │    │ ACI      │    │ App Ins. │
│ Datastores│   │ Designer │    │ AKS      │    │ Data     │
│ Labeling │    │ Notebooks│    │ Edge     │    │ Drift    │
└──────────┘    └──────────┘    └──────────┘    └──────────┘
     │               │               │               │
     └───────────────┴───────────────┴───────────────┘
              Azure ML Workspace
```

7.3 Azure OpenAI Service

Azure OpenAI Service provides REST API access to OpenAI's powerful language models including GPT-4, DALL-E, and Whisper. It includes enterprise features such as private networking, responsible AI content filtering, and regional availability.

Model	Capability
GPT-4o / GPT-4	Large language models for text generation, summarization, code, Q&A, and chat
DALL-E 3	Image generation from text prompts
Whisper	Speech-to-text transcription and translation
Embeddings	Text embeddings for semantic search and similarity
Prompt Engineering	Techniques: zero-shot, few-shot, chain-of-thought, RAG

Chapter 8: Azure DevOps & Governance

Azure DevOps provides developer services for support teams to plan work, collaborate on code development, and build and deploy applications. Azure governance services help organizations maintain control over their cloud environments.

8.1 Azure DevOps Services

Azure DevOps Services Suite

```
┌─────────────────────────────────────────────────────────┐
│                   Azure DevOps                          │
│                                                         │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐  │
│  │  Boards  │ │  Repos   │ │Pipelines │ │ Artifacts│  │
│  │          │ │          │ │          │ │          │  │
│  │ Kanban   │ │ Git repos│ │CI/CD     │ │ Package  │  │
│  │ Backlogs │ │ PRs      │ │ Builds   │ │ Feeds    │  │
│  │ Sprints  │ │ Policies │ │ Releases │ │ NuGet/npm │  │
│  └──────────┘ └──────────┘ └──────────┘ └──────────┘  │
│                  ┌──────────┐                           │
│                  │   Test   │                           │
│                  │   Plans  │                           │
│                  └──────────┘                           │
└─────────────────────────────────────────────────────────┘
```

8.2 CI/CD Pipeline Architecture

Azure DevOps CI/CD Pipeline

```
Code         Build         Test           Deploy
┌─────┐    ┌─────────┐   ┌──────────┐   ┌──────────┐
│ Git │───▶│Compile  │──▶│Unit Tests│──▶│  Dev     │
│ Push│    │Package  │   │Int. Tests│   │  Env     │
│ PR  │    │Analyze  │   │Security  │   └────┬─────┘
└─────┘    └─────────┘   └──────────┘        │
                                        ┌────▼─────┐
                                        │ Staging  │
                                        │ Approval │
                                        └────┬─────┘
                                        ┌────▼─────┐
                                        │Production│
                                        │ Blue/Green│
                                        └──────────┘
```

8.3 Azure Governance

Tool	Purpose
Management Groups	Organize subscriptions into a hierarchy for unified policy and access management
Azure Policy	Create, assign, and manage policies to enforce rules and effects on resources
Azure Blueprints	Define repeatable sets of Azure resources, policies, and role assignments
Resource Tags	Key-value pairs for organizing, searching, and billing allocation of resources
Cost Management	Monitor, allocate, and optimize Azure costs with budgets and alerts
Azure Advisor	Personalized recommendations for reliability, security, performance, and cost

Management Group Hierarchy

```
Root Management Group
└── Corp Management Group
    ├── Production MG
    │   ├── Subscription: Prod-East
    │   └── Subscription: Prod-West
    ├── Non-Production MG
    │   ├── Subscription: Dev
    │   └── Subscription: Test
    └── Sandbox MG
        └── Subscription: Sandbox
```

8.4 Azure Monitor & Observability

Azure Monitor collects, analyzes, and acts on telemetry from Azure and on-premises environments. It maximizes the availability and performance of your applications and services.

Azure Monitor Architecture

```
Data Sources               Azure Monitor
┌─────────────────────┐   ┌─────────────────────────────┐
│ Azure Resources     │──▶│  ┌───────────┐ ┌──────────┐ │
│ Applications (SDK)  │──▶│  │  Metrics  │ │   Logs   │ │
│ Guest OS            │──▶│  │(Time-ser) │ │(Log Anal)│ │
│ Custom Sources      │──▶│  └─────┬─────┘ └────┬─────┘ │
└─────────────────────┘   │        └──────┬─────┘        │
                          │    ┌──────────▼─────────┐    │
```

```
│          │  Insights    │          │
│          │  Alerts      │          │
│          │  Dashboards│          │
│          └──────────────┘          │
└──────────────────────────────────┘
```

Chapter 9: Azure Architecture & Well-Architected Framework

The Azure Well-Architected Framework is a set of guiding tenets that help architects build high-quality, stable, and efficient cloud architectures. It consists of five pillars that represent five areas of architectural consideration for workloads.

Azure Well-Architected Framework – Five Pillars

```
┌───────────────────────────────────────────────────────┐
│         Azure Well-Architected Framework               │
│                                                       │
│   ┌───────────┐ ┌───────────┐ ┌───────────────────┐  │
│   │Reliability│ │ Security  │ │Cost Optimization  │  │
│   │           │ │           │ │                   │  │
│   │ Resiliency│ │ Protect   │ │  Reduce waste     │  │
│   │ Failover  │ │ Detect    │ │  Right-sizing     │  │
│   │ Recovery  │ │ Respond   │ │  Reserved cap.    │  │
│   └───────────┘ └───────────┘ └───────────────────┘  │
│                                                       │
│   ┌────────────────────────┐ ┌──────────────────────┐ │
│   │  Operational           │ │  Performance         │ │
│   │  Excellence            │ │  Efficiency          │ │
│   │  DevOps, Automation    │ │  Scalability         │ │
│   │  IaC, Monitoring       │ │  Caching, CDN        │ │
│   └────────────────────────┘ └──────────────────────┘ │
└───────────────────────────────────────────────────────┘
```

9.1 Common Azure Architecture Patterns

N-Tier Architecture

3-Tier Web Application on Azure

```
Internet
   │
┌──▼────────────────────────────────────────────────────┐
│  Azure Application Gateway + WAF                      │
└──┬────────────────────────────────────────────────────┘
   │            Presentation Tier
┌──▼─────────────────────────────────┐
│  App Service / AKS Web Frontend    │
└──┬─────────────────────────────────┘
   │            Application Tier
┌──▼─────────────────────────────────┐
│  App Service / AKS API Backend     │
└──┬─────────────────────────────────┘
```

```
      │           Data Tier
   ┌──▼────────────────────────────────┐
   │  Azure SQL DB + Azure Cache Redis │
   └───────────────────────────────────┘
```

Microservices Architecture

```
                    Microservices on AKS

   ┌──────────────────────────────────────────────────────┐
   │   API Gateway (Azure API Management)                 │
   └──┬─────────┬─────────┬─────────┬─────────┬───────────┘
      │         │         │         │         │
   ┌──▼──┐   ┌──▼───┐  ┌──▼──┐   ┌──▼──┐   ┌──▼───┐
   │Order│   │User  │  │Inven│   │Pay  │   │Notify│
   │Svc  │   │Svc   │  │tory │   │ment │   │Svc   │
   └──┬──┘   └──────┘  └─────┘   └─────┘   └──▲───┘
      │                                       │
      └──── Azure Service Bus (async events) ─┘
```

9.2 Disaster Recovery Concepts

Concept	Definition	Azure Service
RTO	Recovery Time Objective – max acceptable downtime	Azure Site Recovery
RPO	Recovery Point Objective – max acceptable data loss	Azure Backup
Active-Active	Both sites serve traffic simultaneously	Traffic Manager/Front Door
Active-Passive	Primary serves traffic; secondary is on standby	Azure Site Recovery
Geo-Redundancy	Replicate across regions	GRS/GZRS Storage, SQL Geo-replication

Chapter 10: Azure Pricing, SLAs & Certification Guide

10.1 Azure Pricing Models

Model	Description
Pay-As-You-Go	Pay for compute, storage, and outbound data transfer on a per-second basis
Reserved Instances	1 or 3-year commitment for 40-72% savings on VMs and other services
Spot Instances	Bid on unused Azure capacity for up to 90% savings (evictable)
Azure Hybrid Benefit	Use existing Windows Server/SQL Server licenses on Azure
Dev/Test Pricing	Discounted rates for non-production workloads (requires MSDN subscription)
Free Tier	12 months of popular services free + 55+ always-free services

10.2 Total Cost of Ownership (TCO)

The Azure TCO Calculator helps estimate cost savings of migrating to Azure vs. running on-premises. It considers server, storage, network, and labor costs.

Cost Optimization Best Practices

Right-size resources: Use Azure Advisor recommendations to identify underutilized VMs
Auto-shutdown: Schedule dev/test VMs to shut down outside business hours
Reserved Instances: For steady-state production workloads, 1-3yr reservations save up to 72%
Azure Hybrid Benefit: Apply existing Windows/SQL licenses to save up to 55%
Delete unused resources: Identify and remove orphaned disks, unattached NICs, empty resource groups
Use PaaS over IaaS: PaaS services reduce management overhead and often cost less at scale

10.3 Service Level Agreements (SLAs)

Service	SLA	Compensation
Azure VMs (Avail. Zone)	99.99%	10% → 25% credit
Azure SQL Database	99.99%	10% → 25% credit
Azure Blob Storage	99.9%	10% → 25% credit
Azure Cosmos DB	99.999%	10% → 25% credit
Azure App Service	99.95%	10% → 25% credit
Azure Kubernetes Service	99.95%	10% → 25% credit

Azure Active Directory	99.99%	10% → 25% credit

10.4 Azure Certification Roadmap

```
                    Microsoft Azure Certification Path

  FUNDAMENTALS
  ┌──────────────────┐
  │    AZ-900        │  Azure Fundamentals
  │    AI-900        │  AI Fundamentals
  │    DP-900        │  Data Fundamentals
  └────────┬─────────┘
           │
  ASSOCIATE (Role-Based)
  ┌────────▼──────────────────────────────────────────┐
  │ AZ-104: Azure Administrator                       │
  │ AZ-204: Azure Developer                           │
  │ AZ-500: Security Engineer                         │
  │ DP-203: Data Engineer    │ AI-102: AI Engineer    │
  └────────┬──────────────────────────────────────────┘
           │
  EXPERT
  ┌────────▼──────────────────────────────────────────┐
  │ AZ-305: Solutions Architect Expert                │
  │ AZ-400: DevOps Engineer Expert                    │
  └───────────────────────────────────────────────────┘
```

Exam	Coverage
AZ-900	Azure Fundamentals – Cloud concepts, core services, pricing & SLAs. No prerequisites.
AZ-104	Azure Administrator – Deploy, manage, monitor Azure environments. 6mo experience recommended.
AZ-204	Azure Developer – Design, build, test, maintain cloud applications. Developer background required.
AZ-305	Solutions Architect Expert – Design Azure infrastructure solutions. AZ-104 prerequisite.
AZ-500	Security Engineer – Security controls, threat protection, identity management.
AZ-400	DevOps Engineer Expert – DevOps practices and Microsoft technologies. AZ-104 or AZ-204 required.

10.5 Quick Reference: Azure Service Cheat Sheet

Category	Service	Use When
Compute	Azure VMs	Need full OS control, lift-and-shift
Compute	App Service	Host web apps/APIs without managing VMs
Compute	AKS	Container orchestration at scale
Compute	Azure Functions	Event-driven serverless workloads
Storage	Blob Storage	Unstructured data: images, videos, backups
Storage	Azure Files	SMB/NFS file shares in the cloud
Network	VNet	Private networking for Azure resources
Network	Application Gateway	Layer 7 load balancing + WAF
Network	ExpressRoute	Private dedicated circuit to Azure
Database	Azure SQL	Relational DB, PaaS managed
Database	Cosmos DB	Global, multi-model, low-latency NoSQL
Database	Synapse Analytics	Enterprise data warehouse + big data
AI	Azure OpenAI	GPT-4, DALL-E, Whisper APIs
Security	Key Vault	Secrets, keys, certificate management
Security	Entra ID	Cloud identity and access management
Monitor	Azure Monitor	Metrics, logs, alerts, dashboards
DevOps	Azure DevOps	CI/CD, boards, repos, artifacts

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